MIL-P-20307A (PA)
15 May 1965
SUPERSEDING
MIL-P-20307
21 November 1951

MILITARY SPECIFICATION

POLYVINYL CHLORIDE
(FOR USE IN PYROTECHNICS)

1. SCOPE

1.1 This specification covers one grade of Polyvinyl Chloride intended for use in pyrotechnic items.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-S-48 - Sacks, Shipping, Paper.
RR-S-366 - Sieves, Standard for Testing Purposes.
PPP-D-723 - Drums, Fiber.
PPP-D-729 - Drums, Metal, 55 Gallon (for shipment of noncorrosive material).

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for.
MIL-P-112 - Packaging and Packing for Overseas Shipment Drums, Plywood (for drums whose weight of contents does not exceed 200 pounds).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

FSC: 1370

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MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-650 - Explosives: Sampling, Inspection and Testing.
MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois).

American Trucking Association Publication

National Motor Freight Classification Rules and Container Specifications.

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street, N.W., Washington, D.C.).

3. REQUIREMENTS

3.1 The requirements for Polyvinyl Chloride shall be as specified in Table I when determined as specified in the applicable paragraphs of section 4.4.

TABLE I

Tests	Requirements	Applicable Paragraphs
Color	White	4.4.1
Loss in weight at 135 degrees (C.)	0.5 percent (max.)	4.4.2

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Tests	Requirements	Applicable Paragraphs
Ash	0.1 percent (max.)	4.4.3
Chlorine	55.7 to 57.7 **percent**	4.4.4
Specific viscosity	0.50 to 0.70 **centipoises**	4.4.5
Bulk density	0.40 to 0.60 grams per ml.	4.4.6

3.2 Granulation.-The granulation of Polyvinyl Chloride shall be 95 percent, min. through United States Standard sieve number 100 and 30 percent max. through United States Standard sieve number 325 when determined as specified in 4.4.7.

3.3 Workmanship.-The standard of workmanship shall be such as to insure the production of material meeting the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspection performed.

d. Specification number and date, together with an identification and date of changes.

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e. Statements of findings on all composition and associated ingredients, and certificates of conformance on all remaining material when such material is covered by Government specifications.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure (see 4.1.1) this detail specification, and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with a or b, as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector and, depending on the nature of the deviation, may consist of lot rejection, screening, re-sampling, re-instruction of the supplier's employees, or other appropriate action.

a. Verification at the point of manufacture shall be accomplied at unscheduled intervals in accordance with 4.1.2.1 and 4.1.2.2.

b. Verification at the point of delivery shall be in accordance with 4.1.2.2.

4.1.2.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

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c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.2 Product inspection.-Product inspection shall consist of Government inspection of product which has been previously inspected by the contractor and found to meet the quality assurance provisions of this specification.

4.3 Inspection provisions

4.3.1 Lot formation.-A lot shall consist of one or more batches of polyvinyl chloride, produced by one manufacturer, in accordance with the same specification, same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of polyvinyl chloride that has been subjected to the same unit chemical and physical process intended to make the final lot homogeneous.

Lots shall be submitted in accordance with Standard MIL-STD-105 (or Standard MIL-STD-1235, when applicable.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect.

4.3.2.1 Drums, prior to filling (Polyethylene liner).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Liner cut, torn or punctured....	Visual	01001
Minor:	None defined		

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4.3.2.1.1 Drums, prior to filling (Alternative method).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Interior drum not completely coated	Visual	01002
Minor: None defined			

4.3.2.2 Drums, prior to closing (Polyethylene liner).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Liner not completely sealed	Visual	02001
Minor: None defined			

4.3.2.3 Drum, sealed (Steel).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Closing ring damaged or positioned in such a manner the drum is not closed properly	Visual	03001
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable	Visual	03002

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4.3.2.4 Drum, sealed (Fiber).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Weight of contents (see note)	Scale	04001
102.	Closure incomplete or damaged to the extent that contents sift out	Visual	04002
Minor:	AQL 0.65 percent		
201.	Markings misleading or unidentifiable	Visual	04003

NOTE 1 - The average weight of 10 empty drums will be deducted from the total when determining weight of contents.

4.3.2.4.1 Drum, sealed (Plywood).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Weight of contents	Scale	05001
102.	Strapping missing or broken	Visual	05002
Minor:	AQL 1.00 percent		
201.	Strapping mislocated	Visual	05003
202.	Marking misleading or unidentifiable	Visual	05004

4.3.3 Production testing.

4.3.3.1 Sampling by lot.-A random sample of 10 containers shall be selected from each lot. When lots are comprised of 10 containers or less, each container shall be sampled.

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4.3.3.2 Preparation of composite.-Approximately four ounces primary sample of polyvinyl chloride shall be removed from each of the ten containers in order to equal forty ounces. If there are less than 10 containers, equal primary samples in sufficient quantity to equal forty ounces, shall be removed from each container. The individual primary samples shall then be combined in order to form a homogeneous composite sample of forty ounces and subjected to the tests specified in 4.4. If the composite sample fails to comply with any of the requirements specified, the lot shall be rejected.

4.4 Test methods and procedures.

4.4.1 Color (code No. 06001).-Determine the color of the material by visual inspection.

4.4.2 Loss in weight at 135 degrees C. (code No. 07001).-Loss in weight shall be as determined in MIL-STD-1234 method 102.1 except at 135 degrees C.

4.4.3 Ash (code No. 08001).-Transfer a weighed portion of approximately 2 grams (gms.) of the sample to a tared porcelain crucible. Heat the crucible and contents carefully until the polyvinyl chloride is completely charred and decomposed. Then ignite the crucible and residue at approximately 700 degrees C. until all the carbon is removed, cool in a dessicator and weigh. Calculate the increase in weight as percent ash.

4.4.4 Chlorine (code No. 09001).-Transfer a weighed portion of approximately 0.25 grams (gms.) of the sample to the fusion cup of a Paar bomb of the electric ignition type. Add 0.5 grams (gms.) of finely powdered binzoic acid and mix thoroughly. Add approximately 15 grams (gms.) of sodium peroxide and again mix the contents of the fusion cup thoroughly. Immediately cover the fusion cup with the bomb head which shall have a fuse wire so attached that the wire just touches or is just above the surface of the charge in the cup. Complete the assembling of the bomb. Place the bomb under the contact arm in the water jacket and allow a continuous stream of cold water to pass through the jacket. Ignite the charge using a current of approximately 3 amperes and allow the bomb to cool in the jacket for approximately

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5 minutes. Dismantle the bomb. Rinse the inner surface of the bomb head with hot water and catch the rinsings in a 600 ml. beaker. Place the fusion cup on its side in the beaker and add 100 ml. of water. Cover the beaker with a watch glass and heat the contents of the beaker until decomposition of the sodium peroxide is complete. Remove the cup and rinse with water, catching the rinsings in the beaker. Carefully add 35 percent HNO_3, to the solution in the beaker until the solution is slightly acid. Add a measured portion of 50 ml. of approximately N/10 $AgNO_3$. Heat the contents of the beaker until the precipitate coagulates, and then cool. Filter the solution through a filtering crucible and wash the precipitate with 1 percent HNO_3. Titrate the excess silver nitrate in the combined filtrate and washings with N/10 potassium thiocyanate solution, using 5 to 10 ml. of saturated ferric ammonium sulfate solution as indicator. Also determine the volume of potassium thiocyanate solution required to titrate 50 ml. of silver nitrate solution. Calculate the percent chlorine as follows:

$$\text{Percent chlorine} = \frac{3.346\ (V-v)\ N}{W}$$

where:

V = ml. of potassium thiocyanate solution required to titrate 50 ml. of silver nitrate solution.

v = ml. of potassium thiocyanate solution required to titrate the excess silver nitrate in the determination of the sample.

N = Normality of the potassium thiocyanate solution.

W = gm. of sample

4.4.5 Specific viscosity (code No. 10001).-Transfer a weighed portion of 0.80 grams (gms.) of the sample to a 250 ml. glass-stoppered flash and add 199.2 grams (gms.) of nitrobenzene. Warm the bottle and contents on a steam bath with occasional shaking until the polyvinyl chloride is dissolved. Cool the solution to 20 degrees C. and determine its viscosity in centipoises using a suitable viscosimeter such as a Ubbelohde viscosimeter. Determine the viscosity in centipoises at 20 degrees C. of the nitrobenzene used to dissolve the sample. Calculate the specific viscosity of the Polyvinyl chloride solution as follows:

$$\text{Specific viscosity} = \frac{A}{B} - 1$$

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where:

A = viscosity in centipoises of the solution of polyvinyl chloride in nitrobenzene.
B = viscosity in centipoises of the nitrobenzene.

4.4.6 Bulk density (code No. 11001).-Bulk density shall be as determined in MIL-STD-650 Method 201.1.

4.4.7 Granulation (code No. 12001).-Place a no. 100 U. S. Standard sieve on top of a no. 325 U. S. Standard sieve. Transfer a weighed portion of 25.0 grams of the sample to a 400 ml. beaker containing 175 ml. of isopropyl alcohol, stir the mixture thoroughly and pour it through the assembled sieves. Wash the material through the sieves with a stream of tap water. When no more material appears to pass through the no. 100 sieve, remove this sieve and continue washing the material on the no. 325 sieve until no more passes through. Wet each sieve and the material remaining on it with isopropyl alcohol. Allow the alcohol to drain from the sieves, place the sieves on a steam bath for 5 minutes and dry for 45 minutes in an oven at 135 degrees C. Cool the sieves. Transfer the material remaining on each sieve to a tared weighing dish, weigh the dish and contents and calculate the weight of material retained on each sieve. Calculate the percent passing through each sieve.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Drums.-Unless otherwise specified in the contract or purchase order, polyvinyl chloride shall be packed in a fiber, plywood or metal drum in accordance with Specification PPP-D-723 (Type III, Grade A), MIL-P-112, and PPP-D-729 (Type III, Full Removable Head Type), respectively. Fiber and plywood drums shall be furnished with a nominal 0.004 inch thick polyethylene liner, properly heat sealed or otherwise closed to afford protection to the contents equivalent to that given by a heat seal closure. (Alternately, an equivalent degree of protection may be furnished by suitable laminations or coatings of the inner surface of the drum). The drums shall be closed in accordance with Specification PPP-D-723, PP-D-729 and MIL-P-112 as applicable.

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5.1.1.2 Alternate method.-Polyvinyl chloride may be packed in paper shipping sacks in accordance with Table I, Construction No. 2X of Specification UU-S-48. Closure of sacks shall be in accordance with above Specification.

5.1.2 Level C.-Polyvinyl chloride shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules and Container Specifications for rail shipments or National Motor Freight Rules and Container Specifications for truck shipments, as applicable.

5.2 Marking.-In addition to any special marking required by the contract or purchase order, containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

Custodian:
Army-PA

Preparing activity:
Army-PA

Project No. 1370-A189

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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